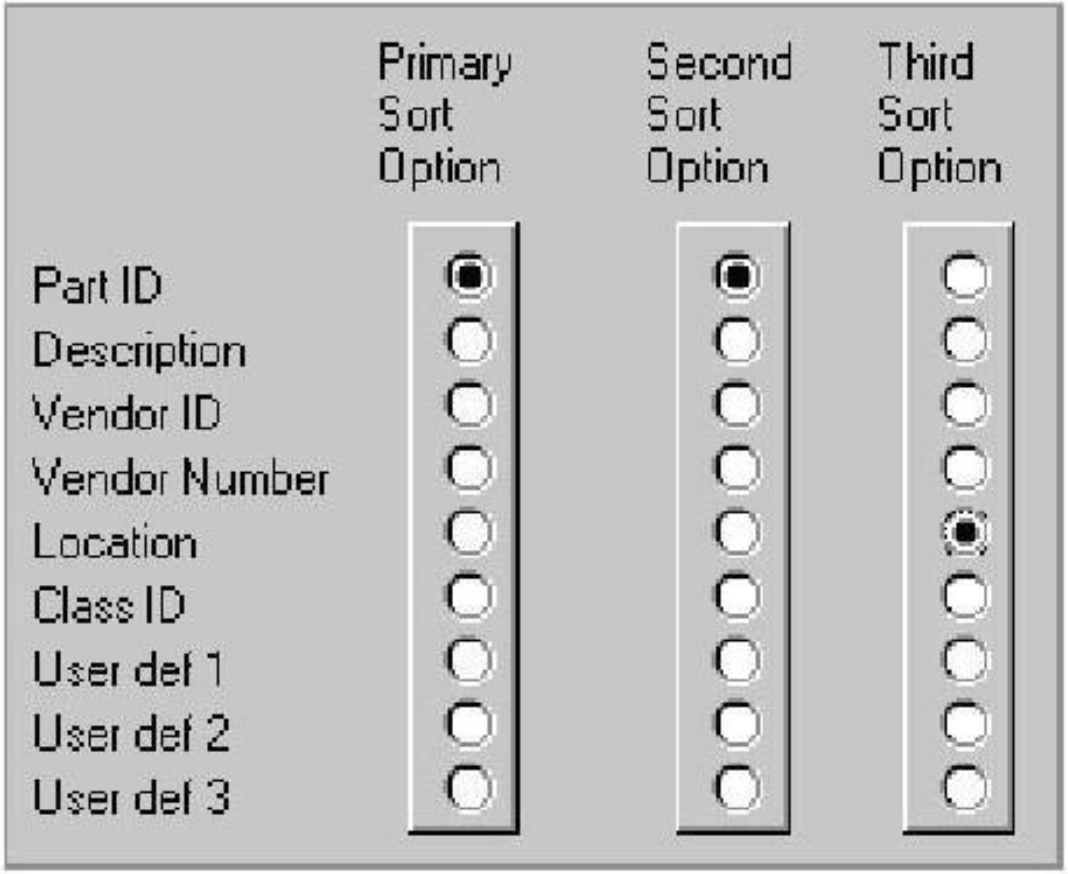
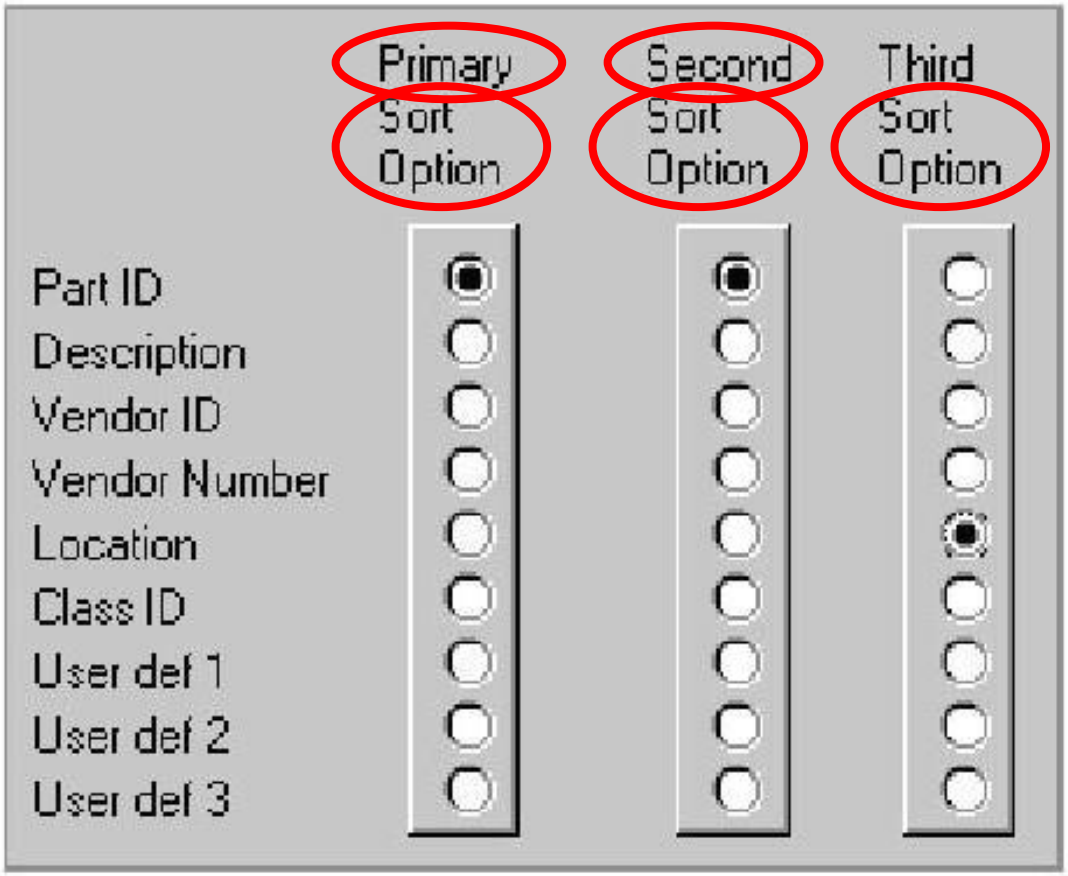
Internationalization

UI Hall of Fame or Shame?



Source: UI Hall of Shame

UI Hall of Fame or Shame?

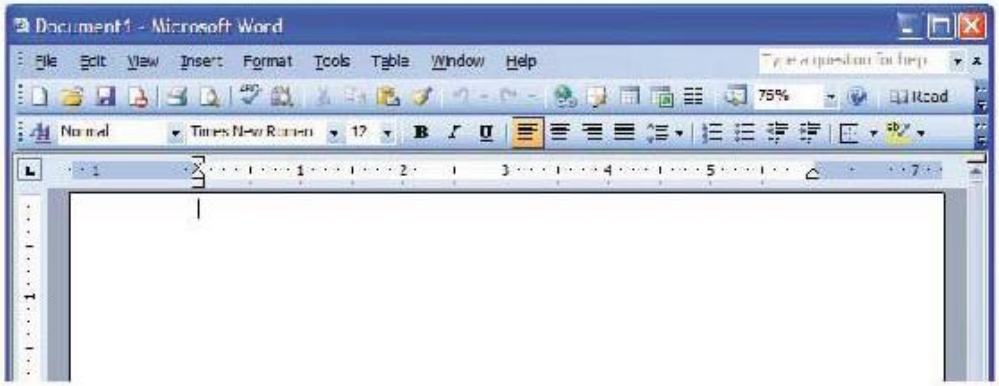


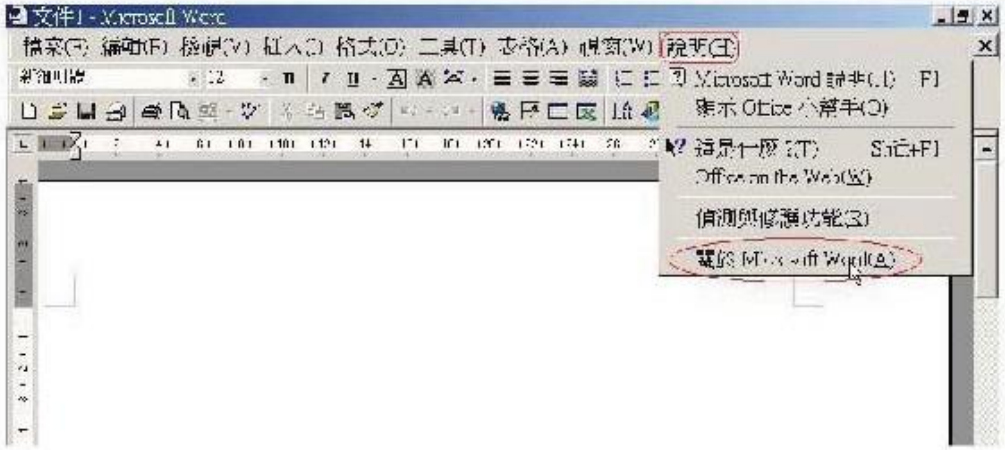
Source: UI Hall of Shame

Today's Topics

- Internationalization
- Design challenges
- Implementation techniques

Internationalization and Localization





The process of making a user interface *ready for translation is called* ***internationalization*** *(often called i18n for* short – 18 because it replaces 18 characters in the middle)

- Doing this translation for a language and culture is called **localization**

Analogy to portability across operating system platforms:

If you write your program carefully so that it doesn't depend on specific features of an operating system or processor, you've made it portable. Making a program portable is analogous to internationalizing it.

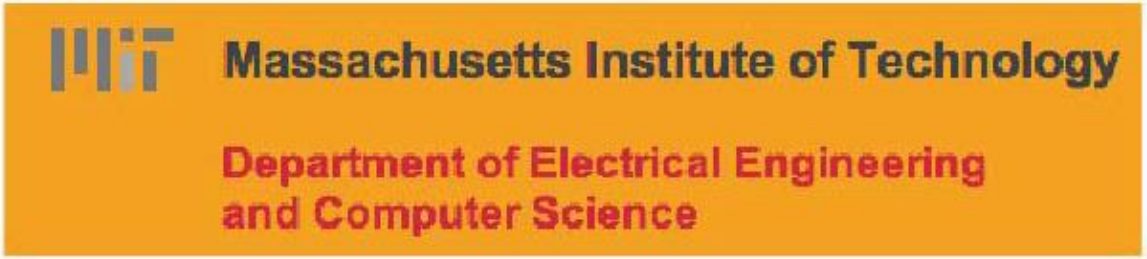
Actually *porting it to another particular platform*, e.g. by recompiling it, is analogous to localizing it.

Translation

- All user-visible text has to be translated
  - Component level

    <button>OK</button>
  - Stroke level

    canvas.fillText("Name:",...)
  - Pixel level



- Error messages too

another reason not to expose internal system names in error messages: An English-reading user might be able to figure out what FileNotFoundException means, but it won't internationalize well.

Risks of Translation

In Wales, official signs are required to be bilingual (English and Welsh).

Welsh actually reads "I am not in the office at the moment. Send any work to be translated. " The translation was outsourced by email

Different Scripts

- Cyrillic

  Все люди рождаются свободными

- Hangul (Korean)

  모든 인간은 태어날 때부터 자유로우며

- Chinese

  人人生而自由，在尊嚴和權利上一律

- Greek

  'Ολοι οι άνθρωποι γεννιούνται

- Arabic

  يولد جميع الناس أحراراً متساوين في الكرامة

distinguish between **script (or alphabet) and language**

Text Direction

- Some scripts don't read left-to-right
  - Arabic, Hebrew are right-to-left
  - Affects drawing, screen layout, even icons

URL box isn't reversed, because it uses the Latin alphabet, not Hebrew. right-to-left languages: when they embed foreign words, or Arabic numbers, the embedded words go in left-to-right order. text might be constantly switching direction

Sort Order

- Unicode order isn't even right for English
- Uppercase/lowercase, accents affect order
- Norwegian: ... x y z æ ø å
- Traditional Spanish: c, ch, d, ..., l, ll, m, ...

U+	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
0000	NUL	SOH	STX	ETX	EOT	ENQ	ACK	BEL	BS	HT	LF	VT	FF	CR	SO	SI
0010	DLE	DC1	DC2	DC3	DC4	NAK	SYN	ETB	CAN	EM	SUB	ESC	FS	GS	RS	US
0020	SP	!	"	#	$	%	&	'	(	)	*	+	,	-	.	/
0030	0	1	2	3	4	5	6	7	8	9	:	;	<	=	>	?
0040	@	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
0050	P	Q	R	S	T	U	V	W	X	Y	Z	[	\	]	^	_
0060	`	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o
0070	p	q	r	s	t	u	v	w	x	y	z	{	|	}	~	DEL
0080	PAD	HOP	BPH	NBH	IND	NEL	SSA	ESA	HTS	HTJ	VTS	PLD	PLU	RI	SS2	SS3
0090	DCS	PU1	PU2	STS	CCH	MW	SPA	EPA	SOS	SGCI	SCI	CSI	ST	OSC	PM	APC
00A0	NBSP	¡	¢	£	¤	¥	¦	§	¨	©	ª	«	¬	SHY	®	¯
00B0	°	±	²	³	´	µ	¶	·	¸	¹	º	»	¼	½	¾	¿
00C0	À	Á	Â	Ã	Ä	Å	Æ	Ç	È	É		Ë	Ì	Í	Î	Ï
00D0	Ð	Ñ	Ò	Ó	Ô	Õ	Ö	×	Ø	Ù	Ú	Û	Ü	Ý	Þ	ß
00E0	à	á	â	ã	ä	å	æ	ç	è	é	ê	ë	ì	í	î	ï
00F0	ð	ñ	ò	ó	ô	õ	ö	÷	ø	ù	ú	û	ü	ý	þ	ÿ

Formatting

- Numbers
  - US/UK 72,350.55
  - France 72 350,55
  - Germany 72.350,55

- Date & time formatting
  - US 10/31/2006 (M/D/Y)
  - Everywhere else 31/10/2006 (D/M/Y)

target for localization therefore needs to be specified by a language/country pair, also called a **locale, such as US English, UK English, or Canadian French**

Color Conventions

US

China funeral

White

Image courtesy of Tamara Evans.

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Red

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Image courtesy of Cormac Heron on Flickr.

wedding

Icons

- Familiar icons in one culture aren't in others

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Implementation Support for I18N

- Message files
- Unicode
- Bidirectionality
- Formatting libraries
- Separating structure from presentation

Message Files

- A message file separates localizable messages from source code
  - Called resource bundles in Java
- Human translators generate a message file for each supported locale
  - Doesn't require translators to read source code or recompile
- Messages with dynamic parts can be tricky
  - **"<N>** users have visited since **<date>"**

Character Sets and Encodings

- Character sets
  - ASCII: A-Z, a-z, 0-9, punctuation, control characters
  - Latin-1: ASCII + accented Latin alphabet
  - Unicode: Latin-1 + Greek, Cyrilic, CJK, math symbols, ...

Chinese, Japanese, and Korean languages

- Fonts map characters to visual appearance

- Encodings map characters to numbers
  - ASCII: A-Z map to 65-90
  - Latin-1: À maps to 192
  - UCS-2: each character maps to 2 bytes
  - UTF-8: each character maps to 1-3 bytes

Bidirectionality

- Bidirectional text display and editing
  - String in memory: This is **arabic text**
  - Drawn on screen:

    (base direction English) This is **txet cibara**

    (base direction Arabic) **txet cibara** This is

- Bidirectional layout
  - FlowLayout goes right-to-left

Formatting Libraries

- Library support for parsing and printing numbers, dates, currency
  - NumberFormatter
  - DateFormatter

Separating Structure From Presentation

- Replaceable icons and images
- Fonts
- Colors

Summary

- Internationalization abstracts a user interface so that it can be localized for different locales
  - Languages
  - Scripts
  - Formatting conventions
  - Cultures